11.7.2 Ringing or girdling experiment

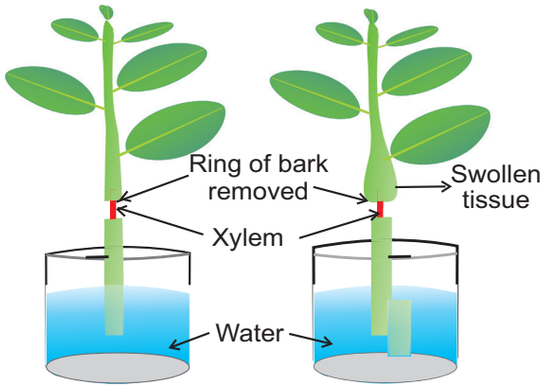


Figure 11.20: Ringing experiment

The experiment involves the removal of all the tissue outside to vascular cambium (bark, cortex, and phloem) in woody stems except xylem. Xylem is the only remaining tissue in the girdled area which connects upper and lower part of the plant. This setup is placed in a beaker of water. After some time, it is observed that a swelling on the upper part of the ring appears as a result of the accumulation of food material (Figure 11.20). If the experiment continues within days, the roots die first. It is because, the supply of food material to the root is cut down by the removal of phloem. The roots cannot synthesize their food and so they die first. As the roots gradually die the upper part (stem), which depends on root for the ascent of sap, will ultimately die.

11.7.3 Direction of Translocation

Phloem translocates the products of photosynthesis from leaves to the area of growth and storage, in the following directions,

Downward direction: From leaves to stem and roots.

Upward direction: From leaves to developing buds, flowers, fruits for consumption and storage. Germination of seeds is also a good example of upward translocation.

Radial direction: From cells of pith to cortex and epidermis, the food materials are radially translocated.

11.7.4 Source and Sink

Source is defined as any organ in plants which are capable of exporting food materials to the areas of metabolism or to the areas of storage. Examples: Mature leaves, germinating seeds.

Sink is defined as any organ in plants which receives food from source.Example: Roots, tubers, developing fruits and immature leaves (Figure 11.21).

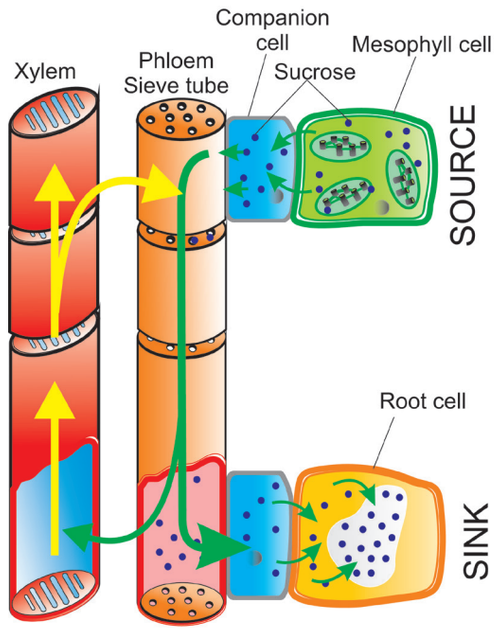


Figure 11.21: Source and Sink

11.7.5 Phloem Loading

The movement of ***photosynthates*** (products of photosynthesis) from mesophyll cells to phloem sieve elements of mature leaves is known as **phloem loading**. It consists of three steps.

i. Sieve tube conducts **sucrose** only. But the *photosynthate* in chloroplast mostly in the form of starch or triose-phosphate which has to be transported to the cytoplasm where it will be converted into sucrose for further translocation.

ii. Sucrose moves from mesophyll to nearby sieve elements by short distance transport.

iii. From sieve tube to sink by long-distance transport.

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